Evaluation of Selected Flavonoids and Antibiotic Combinations Against Biofilms of *Staphylococcus epidermidis* and *Pseudomonas aeruginosa*

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Abbreviations

Abbreviation	Definition
AMR	Antimicrobial resistance
ANOVA	Analysis of variance
AST	Antibiotic susceptibility testing
ATCC	American Type Culture Collection
BGC	Biofilm growth control
CBD	Calgary Biofilm Device
c-di-GMP	Cyclic diguanylate monophosphate
CFU	Colony-forming unit
CV	Crystal violet
DMSO	Dimethyl sulfoxide
eDNA	Extracellular DNA
EPS	Extracellular polymeric substances
FIC	Fractional inhibitory concentration
FICI	Fractional inhibitory concentration index
GC	Growth control
MBC	Minimum bactericidal concentration
MBEC	Minimum biofilm eradication concentration
MHA	Mueller–Hinton agar
MHB	Mueller–Hinton broth
MIC	Minimum inhibitory concentration
MRSA	Methicillin-resistant *Staphylococcus aureus*
MTT	3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
OD	Optical density
ODc	Optical density cut-off value
PA	*Pseudomonas aeruginosa*
PBS	Phosphate-buffered saline
QS	Quorum sensing
SC	Sterility control
SD	Standard deviation
SE	*Staphylococcus epidermidis*
SrtA	Sortase A
WHO	World Health Organization

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Abstract

Wound infections linked to biofilms can be quite stubborn when it comes to antibiotic treatment, and there's growing interest in using plant flavonoids as potential helpers to boost their effectiveness. In this study, we took a closer look at curcumin, baicalein, orientin, and myricetin to see how they fared against Staphylococcus epidermidis ATCC 12721 and Pseudomonas aeruginosa ATCC 10662. We employed a thorough in vitro approach that included disc-diffusion screening, broth microdilution, crystal violet biofilm assays, checkerboard analysis, and a peg-lid challenge to assess viable recovery. Both bacteria were found to be strong biofilm producers. The effectiveness of the flavonoids varied significantly depending on the organism. Baicalein was able to cut down staphylococcal biofilm biomass by 44 to 48% at or above its minimum inhibitory concentration (MIC), while curcumin managed to reduce pseudomonal biomass by 59 to 73% across all tested concentrations. However, curcumin also led to a notable increase in staphylococcal biomass, and orientin didn’t show any measurable impact on S. epidermidis. None of the combinations achieved synergy: vancomycin paired with curcumin was additive (FICI 1.00), while vancomycin with orientin (1.03) and meropenem with curcumin (2.00) showed indifference, even though the MIC for orientin dropped 32-fold when combined. When it came to established biofilms, both combinations managed to suppress optical density and maintain biomass at the highest concentrations, but viable recovery never decreased by more than 0.10 log10. Sterility controls did show growth, which meant the eradication endpoint didn’t meet its validity criteria, and we couldn’t assign a minimum biofilm eradication concentration. Interestingly, three endpoints that are often treated as interchangeable diverged systematically, consistently exaggerating efficacy compared to viable counts. Therefore, any antibiofilm claims based on a single optical endpoint should be approached with caution.

1. Introduction

1.1 Antimicrobial resistance as a global health priority

Antimicrobial resistance (AMR) has evolved over the past ten years from a potential threat to a clear and present danger, now recognized as a significant cause of death. A recent global analysis revealed that bacterial AMR is responsible for over a million deaths each year. If we don't take effective action, it's estimated that around 39 million deaths could be linked to resistant bacterial infections between 2025 and 2050 (GBD 2021 Antimicrobial Resistance Collaborators, 2024). This study is groundbreaking as it places the burden of AMR within a multi-decade context, allowing us to evaluate not just how severe the issue is, but also the trajectory it's on. What stands out is a fundamental mismatch: the pace at which truly innovative antibacterial treatments are being developed and used in clinics simply isn't keeping up with how quickly resistance to current treatments is emerging, becoming established, and spreading.

There are two key takeaways here. First, strategies aimed at prolonging the effectiveness of our current antibiotics hold just as much importance as those focused on developing new ones. This includes adjuvants—compounds that may not be powerful on their own but can enhance the effectiveness of a partner antibiotic. The second takeaway is that the mortality linked to antimicrobial resistance (AMR) isn't spread evenly across different types of infections. A significant portion of this mortality comes from organisms that are not only genetically resistant but also shielded by their growth patterns. This difference between genetic resistance and the tolerance that arises from the physiological state of the population is the core idea behind this work.

1.2 Biofilms as a structural basis of treatment failure

A biofilm is essentially a community of microorganisms that stick to surfaces, all wrapped up in a gooey matrix of substances they produce themselves, known as extracellular polymeric substances (EPS). When these communities lead to chronic infections, they create a unique clinical challenge because they tend to resist both antimicrobial treatments and the body's immune responses, which makes them hard to eliminate and prone to coming back (Ciofu et al., 2022). This matrix isn't just a passive shell; it's made up of polysaccharides, proteins, functional amyloids, lipids, and extracellular DNA (eDNA), along with membrane vesicles. The components of this matrix work together to keep the community stable, help them gather

nutrients, protect the eDNA and exoenzymes, and offer spots for ions and hydrophobic substances to attach (Flemming et al., 2023).

It's important to highlight that final property right from the start, as it plays a crucial role in how we interpret the experiments laid out in this dissertation. If the matrix has sorption sites for hydrophobic molecules, then a candidate antimicrobial that doesn't dissolve well in water might get trapped and concentrated within the EPS, never actually reaching the cells it's meant to target. Many of the flavonoids chosen for this project are quite hydrophobic, which makes the matrix a likely culprit for both diminished effectiveness and skewed optical measurements. By viewing the biofilm as an active physicochemical environment instead of just a passive barrier to diffusion, we can shift our understanding of what a negative result really signifies.

1.3 Wound infection as the clinical context

Wounds offer a fascinating backdrop for exploring why treatments sometimes fail when it comes to biofilms. When the epithelial barrier is compromised, it reveals a nutrient-rich surface that doesn't get enough blood flow, making it an easy target for colonization. At the same time, the lack of proper blood flow means that the levels of antibiotics we can deliver systemically just don't reach the infection site effectively. A review of microbiological data from chronic wound infections around the globe, spanning from 2005 to 2022, showed that Pseudomonas aeruginosa consistently ranked among the most commonly found bacteria across various continents. It also highlighted regional differences in the types of pathogens involved, although biofilm-forming Gram-negative and staphylococcal species remained prominent everywhere (Phan et al., 2023).

Wound infections often involve multiple types of bacteria, and the arrangement within the wound bed isn't just haphazard. In a setting that resembles a chronic wound, you'll find diverse communities of microorganisms working together.

In this study, we focus on using monospecies laboratory reference strains of the two organisms as manageable models. Instead of viewing the wound as just a condition to replicate in the lab, we treat it as the clinical motivation behind our work. This distinction is intentionally upheld throughout the research, as we can't assume that reference strains and isolates derived from wounds will behave in the same way.

1.4 Pseudomonas aeruginosa as a wound pathogen

Pseudomonas aeruginosa is a Gram-negative opportunistic pathogen that poses a significant threat in various clinical settings. It can lead to infections in individuals with cystic fibrosis, burn injuries, weakened immune systems, chronic obstructive pulmonary disease, cancer, and those on mechanical ventilation. What makes this bacterium particularly dangerous is its multifaceted virulence, which isn't due to just one factor. Instead, it relies on a combination of elements like lipopolysaccharide, quorum sensing (QS) systems, two-component regulatory systems, six different secretion systems, outer membrane vesicles, and CRISPR-Cas-associated regulation (Qin et al., 2022). This wide-ranging regulatory capability allows Pseudomonas aeruginosa to quickly adapt to challenging environments, especially those created by antibiotic treatments.

The importance of biofilm formation in this species is well established, especially in the context of wounds and burns, where biofilm production often goes hand in hand with resistance to treatment. In a study involving 40 P. aeruginosa isolates taken from hospitalized burn patients, every single isolate was found to produce biofilm. Alarmingly, 40% of these isolates showed carbapenem resistance, with *bla*TEM, *bla*VIM, and *bla*CTX-M being the most commonly identified β-lactamase genes (Ghasemian et al., 2023). This combination poses a serious challenge: carbapenem resistance strips away a key treatment option, while the biofilm itself diminishes the effectiveness of any remaining therapies. It's this interplay between the two factors, rather than either one alone, that makes exploring additional treatment strategies particularly appealing for this organism.

1.5 Staphylococcus epidermidis as an opportunistic biofilm former

Staphylococcus epidermidis finds itself in a bit of a gray area. As one of the most prevalent members of our skin's microbiota, it doesn't just sit there passively; it actively helps maintain skin health by kickstarting the local immune response and offering a defense against more harmful pathogens. Instead of viewing its potential for causing disease as a simple yes-or-no situation, it's better to think of it as a spectrum, largely due to the significant differences between strains. Despite this complexity, it remains the leading cause of infections related to implants and is well-known for forming opportunistic biofilms (Severn and Horswill, 2023).

When it comes to chronic wounds, the microenvironment seems to favor the more troublesome side of things. A study on S. epidermidis populations and their resistome in venous leg ulcer tissue revealed that patients had multidrug-resistant, biofilm-forming strains. These isolates not only formed biofilms but also triggered a pro-inflammatory response in a human wound model,

which was linked to poor healing outcomes (Dinić et al., 2024). This shows that a normally harmless organism can pick up resistance and biofilm-forming abilities in wound conditions, ultimately hindering the healing process.

This dual nature has some important implications. Since S. epidermidis is a regular inhabitant of our skin, finding it in a wound doesn't automatically mean it's causing an infection. Plus, just because it shows antibiofilm activity against a lab reference strain doesn't mean we can apply those findings to the entire species.

1.6 Mechanisms underlying biofilm antimicrobial tolerance

The decreased ability of biofilms to respond to antimicrobials is due to a variety of factors, and there isn't just one reason behind it (Ciofu et al., 2022). Typically, we identify four main contributors, and each one has unique implications for how we design adjuvants.

Restricted penetration is probably the easiest concept to grasp. The EPS slows down the diffusion of certain antimicrobials and can even bind to others through electrostatic or hydrophobic interactions (Flemming et al., 2023). This creates a concentration gradient throughout the biofilm, meaning that cells located deeper within may never come across an inhibitory concentration, even if the overall concentration is higher than the minimum inhibitory concentration (MIC) for planktonic cells.

Metabolic diversity adds another layer to the complexity. Variations in oxygen and nutrient levels create distinct subpopulations within a community. Since many antibiotics rely on active cellular processes, cells that are slow-growing or in a dormant state tend to be less vulnerable, regardless of their genetic makeup. These persister-like groups can endure antibiotic exposure without developing permanent resistance, allowing them to bounce back and repopulate the community. This is a key reason why regrowth after what seems like successful treatment is a common feature in biofilm infections.

When you put all these mechanisms together, it becomes clear why a compound that can reduce a planktonic MIC might not work at all against a fully developed biofilm. This also suggests that the best experimental designs are the ones that look at multiple endpoints.

1.7 Inhibition, biomass reduction and eradication are not equivalent endpoints

One common issue in the antibiofilm research is the tendency to mix up endpoints that are actually quite different both conceptually and operationally. It's crucial to distinguish between three specific endpoints, especially since this study examines all three and finds that they don't align.

To assess the prevention of biofilm formation, researchers typically expose cells to a compound during the attachment and maturation stages and then measure the biomass that builds up. The retained biomass is usually quantified using crystal violet staining, with results compared against an optical density cut-off (ODc) established from uninoculated controls (Stepanović et al., 2000). However, it’s important to note that crystal violet is a biomass stain, not a viability stain. It binds to both dead cells and the matrix just as easily as it does to living cells, meaning that a decrease in staining indicates less material overall, not necessarily fewer living organisms.

On the other hand, eradicating an established biofilm is a different challenge and is the endpoint that holds the most clinical significance, as interventions typically happen after a biofilm has already formed. The Calgary Biofilm Device tackled this issue by growing 96 identical biofilms on a peg lid that can be transferred to a challenge plate, allowing for the determination of minimum inhibitory concentration (MIC) and minimum biofilm eradication concentration (MBEC) values simultaneously (Ceri et al., 1999). The MBEC is defined as the lowest concentration that successfully eradicates the biofilm, which is confirmed by observing clear wells after recovery and sonication (Innovotech Inc., 2025).

In between these two endpoints lies metabolic suppression, which this study assesses using tetrazolium (MTT) reduction. This method measures the cellular reductive capacity, which can decrease without the population actually being killed. Therefore, these three endpoints can diverge in meaningful ways: a biofilm might still hold onto its matrix even after losing metabolic activity, or it could maintain a small viable population despite a significant loss in biomass. Only reporting the endpoint that looks most favorable would give a skewed picture of the underlying biology.

1.8 Flavonoids as candidate antibiotic adjuvants

Flavonoids are among the largest groups of plant secondary metabolites. They’re produced partly as a response to microbial threats and work through various mechanisms instead of targeting just one specific pathway. Interestingly, some of them seem to selectively impact bacterial cells rather than those of the host (Górniak et al., 2019). However, their appeal is

somewhat diminished by a couple of drawbacks: their intrinsic potency tends to be on the lower side, with minimum inhibitory concentration (MIC) values often reaching into the hundreds of micrograms per milliliter, and they don't dissolve well in water. This poor solubility means that we often need to use cosolvents, which can also have antibacterial properties.

These limitations are precisely why considering it as an adjuvant is a more effective strategy than treating it as a monotherapy. A compound that may not work well on its own as an antibacterial could still be incredibly valuable if it helps lower the dosage of a traditional agent needed for effectiveness, disrupts the formation of the matrix, or interferes with the regulatory systems that maintain the biofilm. Additionally, anti-virulence strategies are especially attractive because they don't exert the same intense selective pressure for resistance that growth inhibition often leads to.

One important point that often gets overlooked in this discussion is that we really need to show adjuvant activity against the specific endpoint that truly matters. Just reducing a QS-regulated phenotype, cutting down on matrix staining, and eliminating viable cells are all distinct claims, and each one needs its own set of evidence to back it up.

1.9 Rationale for selecting curcumin, baicalein, orientin and myricetin

We chose four different compounds to represent a variety of proposed mechanisms and specificities for different organisms. This way, the screening process would focus on testing a range of mechanisms instead of just variations of one particular mode of action.

Curcumin

Curcumin, a diarylheptanoid polyphenol derived from Curcuma longa, has shown impressive anti-inflammatory and anti-virulence properties. In tests against a Bap-expressing strain of S. aureus, it achieved the lowest minimum inhibitory concentration (MIC) among the polyphenols evaluated, coming in at 375 μg/mL (Matilla-Cuenca et al., 2020). While this is quite potent compared to other flavonoids, it's still much higher than what can be realistically achieved in the body. Additionally, curcumin is known for its vibrant color and poor solubility in water, which makes it both a highly promising compound and a tricky one to work with in the lab. These characteristics can interfere with optical measurements, regardless of any actual biological effects.

Baicalein

Baicalein is a flavone that plays a role in disrupting the quorum sensing of P. aeruginosa. When used at sub-MIC concentrations, it reduces various QS-controlled virulence traits, such as motility and biofilm formation. It also impacts secreted factors like pyocyanin, LasA protease, LasB elastase, and rhamnolipids. Additionally, baicalein has been shown to dampen the pro-inflammatory response of macrophages to infection by affecting the MAPK and NF-κB pathways (Luo et al., 2016). More recent studies have looked at baicalein in combination with quercetin and azithromycin against P. aeruginosa PAO1, focusing on biofilm formation, elastase, pyocyanin, and rhamnolipid production, as well as the expression of the lasR and mvfR regulatory genes (Parra Rodríguez et al., 2026). As a result, baicalein emerged as the primary candidate for the panel targeting quorum sensing.

Orientin

Orientin is a C-glycosylated flavone that was chosen for its well-defined molecular target in Gram-positive bacteria. It effectively inhibits S. aureus sortase A (SrtA) with a half-maximal inhibitory concentration of 50.44 ± 0.51 μM. This inhibition leads to a reduction in fibrinogen binding, less biofilm formation, and a decrease in the surface display of staphylococcal protein A. The binding occurs at specific residues: Glu-105, Thr-93, and Cys-184. Notably, treatment with orientin has been shown to protect mice from lethal MRSA pneumonia (Wang et al., 2021). Since SrtA plays a crucial role in anchoring surface proteins to the Gram-positive cell wall and is key to adhesion, it's an important pathway for staphylococcal biofilm formation. However, it remains to be seen whether this relevance extends from S. aureus to S. epidermidis, given their differences in surface protein profiles and matrix composition. This study aims to explore that question.

Myricetin

Myricetin is a flavonol that has shown activity against both types of target organisms, which is one of the reasons it was chosen for this study. In staphylococci, it was found to inhibit biofilm formation that relies on the biofilm-associated protein Bap by blocking the assembly of Bap-related amyloid-like aggregates. This action also led to a reduction in catheter colonization in an in vivo model (Matilla-Cuenca et al., 2020). When it comes to P. aeruginosa, myricetin enhanced the effectiveness of azithromycin, ciprofloxacin, and cefdinir, while also decreasing the synthesis of cyclic diguanylate monophosphate (c-di-GMP) by 28%, 57%, and 30%, respectively. This reduction helped suppress biofilm formation and hindered swimming, swarming, and twitching motility by interfering with FimX-mediated c-di-GMP signaling

(Zeng et al., 2025). It's important to note that this last study specifically illustrates antibiotic potentiation rather than intrinsic activity, making it the most relevant published example for the adjuvant hypothesis being tested here.

1.10 Quantifying antimicrobial interaction and the limits of the FICI

When it comes to understanding how two antimicrobials interact, we often turn to a method called checkerboard microdilution. This technique gives us a value known as the fractional inhibitory concentration index (FICI), which is calculated by taking the sum of each drug's minimum inhibitory concentration (MIC) when used together and dividing it by the MIC of each drug on its own. According to the widely accepted guidelines, an FICI of 0.5 or lower suggests that the two drugs work well together (synergy), values between 0.5 and 4 indicate that they don’t really affect each other, and anything above 4 suggests they might actually hinder each other (antagonism) (Odds, 2003). However, while the calculation seems straightforward, there are some important limitations to keep in mind regarding the results we get.

First off, the FICI is based on growth inhibition measured in a planktonic culture, which means it reflects the concentration needed to stop free-floating cells. This doesn’t necessarily translate to what happens in a biofilm, where factors like diffusion, metabolic differences, and persister cells come into play. Secondly, since the FICI is calculated from two-fold dilution series, it has a somewhat rough resolution; just a small measurement error can shift a combination from one category to another. Lastly, the thresholds we use are based on convention rather than any underlying mechanism, and there are various published standards, so it’s crucial to clearly state which criteria we’re using if we want our results to be comparable.

These limitations don’t make the checkerboard method useless, but they do shape how we should use it. In this study, we’re treating the FICI as a screening tool, using it alongside data from single-agent biofilm tests to help us prioritize which combinations to test on established biofilms, rather than relying on it to predict how those biofilms will behave.

1.11 Research gap addressed by the present study

The literature we've looked at is quite broad, but it has some significant gaps. While we've delved deeply into the mechanisms of individual flavonoids, this research often focuses on just one organism at a time, at a specific stage of experimentation. Plus, the conditions vary widely from study to study, including differences in strain, medium, solvent, timing of exposure, and

the methods used to measure outcomes. For instance, orientin has been studied down to the level of its individual binding residues in

It's still uncertain if these findings would hold up when evaluated under a single, consistent protocol. Specifically, we don't know if the reported decreases in biomass or virulence actually relate to a reduction in recoverable viable cells from an established biofilm. To date, there doesn't seem to be any published research that looks at all four compounds together, testing them against both a Gram-positive and a Gram-negative organism, and following a design that spans from susceptibility screening to peg-lid eradication and viable recovery endpoints.

The main takeaway from this work is the sequence itself. What's really important here isn't just pinpointing one effective compound, but rather examining whether the endpoints used in existing literature align when we apply them to the same organisms, compounds, and conditions. It's also crucial to be transparent about the discrepancies when they arise.

1.12 Hypothesis, aim and objectives

We *hypothesize* that curcumin, baicalein, orientin, and myricetin will show significant antibacterial and antibiofilm properties against S. epidermidis and P. aeruginosa. Furthermore, we believe that combining these compounds with traditional antibiotics will boost their antimicrobial effectiveness compared to using each agent on its own, even against established biofilms.

The goal of this study is to assess the antibacterial and antibiofilm properties of four chosen flavonoids, both on their own and when paired with a standard antibiotic, against lab reference strains of S. epidermidis and P. aeruginosa. We also want to find out if the combinations that show promise in preventing planktonic growth and biofilm formation can actually suppress or eliminate an already established biofilm.

Objectives. The specific objectives of the study were as follows.

1. To characterise the antibiotic susceptibility profile of both organisms by disc-diffusion screening, and to select an appropriate conventional antibiotic partner for each.
2. To determine MIC values for the selected antibiotics and for each of the four flavonoids by broth microdilution.
3. To classify the baseline biofilm-forming capacity of both organisms by crystal violet staining against an ODc-derived threshold.

4. To quantify the effect of each flavonoid on biofilm biomass across a concentration series relative to untreated controls.

5. To assess antibiotic–flavonoid interactions by checkerboard microdilution and to classify them using the FICI under a prespecified criterion.

6. To evaluate the selected combinations against established biofilms using growth-associated optical density, tetrazolium reduction and crystal violet biomass endpoints.

7. To determine viable recovery after antimicrobial challenge, to calculate log10 reductions relative to untreated control pegs, and to assess whether a protocol-valid MBEC can be assigned within the concentration range tested.

Objectives 1 to 5 establish the screening basis for combination selection; objectives 6 and 7 constitute the critical test of the hypothesis, since they alone distinguish suppression of a measurable signal from eradication of a viable population.

2. Materials and Methods

2.1 Study design and experimental workflow

We carried out an in vitro study using pure cultures of Staphylococcus epidermidis ATCC 12721 and Pseudomonas aeruginosa ATCC 10662, both sourced from the university's stock collection and examined individually. The study followed a sequential design, where each step provided the necessary parameters for the next (see Table 1): we began with screening to select a comparator antibiotic for each organism, then used microdilution to determine the MIC values. Next, we employed crystal violet staining to assess biofilm capacity, followed by an MIC-derived series to measure the impact on biofilm formation. We then classified interactions through checkerboard microdilution, and the combinations that showed promise were taken forward to the peg-lid challenge and recovery phase.

Table 1. This is a sequential experimental workflow that outlines the purpose and main endpoint measured at each stage. Each stage provided the necessary parameters to inform the design of the next one, ensuring that the selection of compounds and combinations was based on actual data rather than just assumptions made beforehand.

Stage	Assay	Purpose	Primary endpoint
1	Disc-diffusion screening	Screen antibiotic response; select a comparator antibiotic per organism	Zone of inhibition (mm)
2	Broth microdilution	Determine MIC for antibiotics and flavonoids	MIC (μg/mL)
3	Crystal violet microtitre assay	Confirm and classify biofilm-forming capacity	OD550 relative to ODc
4	Biofilm inhibition series	Quantify change in biofilm biomass across concentrations	OD550; % change vs control
5	Checkerboard microdilution	Classify antibiotic–flavonoid interaction	FICI
6	Peg-lid challenge	Assess growth, metabolic activity and retained biomass of established biofilm	OD650; MTT; OD570
7	Recovery and viable enumeration	Determine surviving viable cells and assess eradication	Log10 CFU/peg; log10 reduction

2.2 Bacterial strains, media and reagents

You can find a summary of the materials and instrumentation in Table 2. We consistently used Mueller–Hinton agar and broth to keep the growth medium uniform throughout the study.

Table 2. This includes the main biological materials, media, reagents, consumables, and instruments, along with how each one is used throughout the experimental process.

Category	Item	Application
Bacterial strains	*Staphylococcus epidermidis* ATCC 12721; *Pseudomonas aeruginosa* ATCC 10662	Test organisms
Flavonoids	Curcumin, baicalein, orientin, myricetin	Natural-product test agents

Category	Item	Application
Antibiotics	Vancomycin (*S. epidermidis*); meropenem (*P. aeruginosa*)	Comparator and combination partners
Media	Mueller–Hinton agar; Mueller–Hinton broth	Susceptibility testing, growth, biofilm culture
Buffers and solvents	0.9% w/v saline; phosphate-buffered saline; dimethyl sulfoxide; ethanol	Suspension, washing, solubilisation, dye elution
Stains and indicators	0.01% w/v crystal violet; MTT	Biomass quantification; metabolic activity
Consumables	96-well and 24-well flat-bottomed plates; Calgary-type peg lid; antibiotic discs	Microdilution, biofilm culture, screening
Instrumentation	Microplate spectrophotometer (OMEGA); shaking incubator; ultrasonic bath	Optical density; culture; biofilm recovery

2.3 Inoculum standardisation

We used two different standardization procedures, and we're reporting them separately since they aren't equivalent. For the microdilution method, we prepared a suspension of about 5×10^5 CFU/mL by doing a serial dilution in sterile saline across four tubes (labeled a–d) that contained 1, 10, 9, and 19 mL, respectively. We swabbed colonies into tube a, then transferred 100 µL from a to b, followed by 1 mL from b to c, and finally, 1 mL from c to d.

For the biofilm assays, we started by growing colonies in 10 mL of Mueller–Hinton broth inside a shaking incubator for 24 hours. After that, we normalized the cultures in phosphate-buffered saline to an optical density of $OD_{600} = 0.5$ using the formula $V_1 = (C_2V_2)/C_1$, where C_1 represents the overnight optical density. Meanwhile, the peg-lid cultures were adjusted to an OD_{600} of 0.1, allowing for growth during the extended incubation period used in that setup.

2.4 Antibiotic susceptibility testing

We carried out a disc diffusion test using Mueller–Hinton agar. First, we spread standardized suspensions evenly over the surface, then placed the discs according to the potencies listed in Table 3. The plates were incubated at 30 °C for 24 hours, after which we measured the zones of inhibition. For P. aeruginosa, we screened eight agents in triplicate, while for S. epidermidis, we did so in duplicate.

Incubating at 30 °C is a bit off from the standard reference range of 35 ± 2 °C. As a result, we're reporting the zone diameters descriptively, without assigning any susceptibility categories; the screen was just used to pick a comparator antibiotic. For S. epidermidis, we chose vancomycin, and for P. aeruginosa, we went with meropenem. This deviation is discussed further in Section 4.7.

Table 3. We tested antibiotic discs on each organism, noting their codes and strengths. The panels varied for each organism to account for their unique susceptibility. Tetracycline and colistin sulphate were used at different strengths for the two organisms.

Antibiotic	**Code**	**Potency, *P. aeruginosa* (µg)**	**Potency, *S. epidermidis* (µg)**
Penicillin	P	2	2
Gentamicin	CN	10	10
Amoxycillin	AMC	30	Not tested
Tetracycline	TE	10	30
Chloramphenicol	C	30	Not tested
Sulfamethoxazole	SXT	25	25
Meropenem	MEM	10	Not tested
Colistin sulphate	CT	20	10
Vancomycin	VA	Not tested	5
Erythromycin	E	Not tested	15
Linezolid	LZD	Not tested	10

2.5 Flavonoid preparation and MIC determination

Since 100% dimethyl sulfoxide is effective at killing bacteria, we prepared our stocks at eight times the working concentration and then diluted them into broth, which cut down the final solvent exposure by a factor of eight. For a 1024 µg/mL solution, we needed 8.192 mg of the compound in 1 mL of dimethyl sulfoxide, and we mixed 125 µL of that with 875 µL of broth. We tested orientin, curcumin, and baicalein starting from 1024 µg/mL, while myricetin was tested from a concentration of 100 µg/mL.

We determined the MIC using a two-fold broth microdilution method in 96-well plates, set up as shown in Table 5. The plates were incubated at 30 °C for 24 hours. After that, we visually assessed the growth and then added 20 µL of MTT, allowing for an additional 20 to 30 minutes of incubation. Finally, we recorded the MIC based on the criteria outlined in Table 6.

2.6 Baseline biofilm formation and single-flavonoid inhibition

We measured biofilm formation using crystal violet staining in 24-well plates, following the setup outlined in Table 5, and incubated them for 24 hours at 37 °C. After that, we carefully inverted the plates to discard the spent medium and rinsed the wells twice with phosphate-buffered saline. To stain the adherent biomass, we applied 0.01% w/v crystal violet for 15 minutes at 37 °C or for an hour at room temperature. We then washed the wells until the wash liquid ran clear. The retained dye was dissolved in 200 µL of ethanol, transferred to a 96-well

plate, and measured at 550 nm. The capacity was then classified based on the optical density cut-off shown in Table 6.

The inhibition experiments were conducted in the same way, but this time we added flavonoid treatment during the entire biofilm formation process at concentrations of 2X, X, X/2, and X/4 relative to the minimum inhibitory concentration (MIC) for each compound and organism (see Table 4). To determine the percentage change, we used the formula [(mean control OD − mean treated OD)/mean control OD] × 100, comparing it to the untreated control from the same plate. If the values were negative, it indicated that the treated biomass was greater than the control, and we reported those values as they were instead of rounding them down to zero.

Table 4. Absolute flavonoid concentrations corresponding to the relative series used in the biofilm inhibition assays, derived from the MIC determined for each compound against each organism. Orientin and myricetin were not carried forward against *P. aeruginosa*, as no MIC was reached within the range tested.

Organism and flavonoid	2X (µg/mL)	X (µg/mL)	X/2 (µg/mL)	X/4 (µg/mL)
S. epidermidis — baicalein	1024	512	256	128
S. epidermidis — curcumin	128	64	32	16
S. epidermidis — orientin	2048	1024	512	256
S. epidermidis — myricetin	100	50	25	12.5
P. aeruginosa — baicalein	512	256	128	64
P. aeruginosa — curcumin	512	256	128	64

2.7 Checkerboard assay and FICI

We evaluated the interaction using a checkerboard microdilution method, setting up paired plates as shown in Table 5. On plate A, we created an antibiotic gradient, while plate B featured a gradient of flavonoids. We transferred 50 µL from each well of plate B into the matching wells of plate A, moving from column 12 back to column 2. After that, we added 100 µL of inoculum to each well. In the highest wells, both agents were introduced at double their minimum inhibitory concentration (MIC).

The fractional inhibitory concentration (FIC) for each agent was calculated by taking its MIC when used in combination and dividing it by its MIC when used alone. The fractional inhibitory concentration index (FICI) was determined by summing these values. We used the interpretive bands outlined in Table 6, applying them exactly as specified without any changes.

2.8 Established-biofilm challenge and recovery

We put the retained combinations to the test against established biofilms using a Calgary-type peg-lid device, following the principles of minimum biofilm eradication concentration (Ceri et al., 1999; Innovotech Inc., 2025). We normalized the cultures to an OD_{600} of 0.1 and dispensed 150 µL into each well beneath the peg lid, then placed them in a shaking incubator to allow for biofilm development. Meanwhile, we prepared a challenge plate featuring a two-fold combination gradient, along with controls for sterility, growth, and biofilm growth (see Table 5), before transferring the peg lid into the plate.

The lid was then moved to a recovery plate (see Table 5) and sonicated for 30 minutes to help release any cells that were sticking around. From the recovery plate, we took 100 µL and added it to fresh medium for measurement at 650 nm, while 10 µL was spotted onto agar. For the challenge plate, we diluted 20 µL into 180 µL of phosphate-buffered saline and spotted 10 µL of the leftover solution onto agar. We also evaluated the challenge wells using MTT and crystal violet, measuring at 570 nm.

To calculate the viable density, we used the formula $Log_{10}(CFU/peg) = log_{10}[((X/B) \times D \times V) + 1]$. Here, X represents the colony count, B is the spotted volume (0.01 mL), V stands for the well volume (0.20 mL), and D is the dilution factor, which is set to 10 for challenge counts and 1 for recovery counts. The Log_{10} reduction is determined by subtracting the density of the treated pegs from the mean control density. You can find the criteria for MBEC and sterility control in Table 6.

Table 5. Plate configurations and control wells for the microdilution, checkerboard, challenge and recovery assays. Volumes and transfer sequences are given as performed.

Assay	**Plate configuration**	**Control wells**
Broth microdilution (MIC)	Column 1 left empty; 100 µL broth into columns 2–12; 200 µL test agent into column 1; 100 µL transferred serially from column 1 to column 11, excess discarded; 100 µL inoculum into all wells except column 12	Column 12: broth only, no agent and no inoculum (sterility control)
Checkerboard, plate A (antibiotic)	50 µL broth into all wells except B2–H2; 100 µL antibiotic into B2–H2, diluted serially left to right in 50 µL steps to column 12; row A prepared likewise from A2	A1: broth and inoculum only (growth control)
Checkerboard, plate B (flavonoid)	100 µL broth into C2–H12; 200 µL flavonoid into B2–B12, diluted serially downwards in 100 µL steps to row H; 50 µL then transferred from each well of plate B into plate A, column 12 back to column 2	Carried on plate A
Peg-lid challenge	200 µL combination at twice the highest intended concentration into row A; 100 µL broth into rows B–H; 100 µL added to rows B and C; two-fold serial dilution from row C to row H, excess discarded; further 100 µL broth into rows C–H	Sterility control: highest concentration, no inoculum. Growth control and biofilm growth control: broth, no antimicrobial

Assay	Plate configuration	Control wells
Recovery	200 µL phosphate-buffered saline per well; peg lid transferred in, sealed and sonicated for 30 min	Recovery controls carried from the corresponding challenge wells
Crystal violet biofilm assay (24-well)	1 mL Mueller–Hinton broth and 100 µL inoculum normalised to OD600 = 0.5 per well, in triplicate; incubated 24 h at 37 °C	Three uninoculated wells per plate: medium only, serving as blanks and as contamination indicators

2.9 Data handling and statistical analysis

We calculated percentages and $\log_{10}$ values directly from the raw replicate readings instead of using rounded or graphed figures, and all analyses were conducted using Minitab. To check for residual normality, we used the Anderson–Darling test, while Levene's test helped us assess variance homogeneity. When both conditions were met, we applied one-way analysis of variance with Dunnett's comparisons against the control. If the variances were unequal, we turned to Welch's analysis of variance with Games–Howell. For non-normal multiple-group data, we used the Kruskal–Wallis test, and for baseline comparisons, the Mann–Whitney test was employed. We set the significance level at $p < 0.05$, and any values shown as 0.000 are actually reported as $p < 0.001$.

It's important to note one limitation regarding our inferences. The replicates were prepared in triplicate on a single plate, and we didn't repeat any experiments on different days with independently prepared cultures; thus, these are technical replicates rather than biological ones. Since inferential tests assume that experimental units are independent, the statistics presented in Chapter 3 reflect variation between wells within a plate and shouldn't be generalized to variations across different experiments. We've included the direction and magnitude of effects with each test, and we discuss this limitation further in Section 4.7. The interpretations for MIC, FICI, MTT, and MBEC were based on criteria defined by the assay (see Table 6) rather than through inferential testing.

Table 6. We established specific interpretive criteria for each antimicrobial and biofilm endpoint, clearly noting the source for each criterion. These criteria were set in stone prior to the analysis and were used exactly as they were defined, without any changes.

Parameter	Criterion	Basis
MIC	Lowest concentration at which the well remained yellow after MTT addition, indicating no reduction of the tetrazolium salt	Assay-defined threshold
Biofilm classification	Non-adherent, $OD \leq ODc$; weak, $ODc < OD \leq 2 \times ODc$; moderate, $2 \times ODc < OD \leq 4 \times ODc$; strong, $OD > 4 \times ODc$	Stepanović et al. (2000)
ODc	Mean optical density of the negative control plus three standard deviations of the negative control	Stepanović et al. (2000)

Parameter	Criterion	Basis
FICI	Synergistic, ≤ 0.5; additive, = 1; indifferent, > 1 to ≤ 4; antagonistic, ≥ 4	Departmental protocol; consistent with Odds (2003)
MBEC	Lowest concentration at which no growth was evident following recovery	Ceri et al. (1999); Innovotech Inc. (2025)
Sterility control	Wells must remain clear; growth renders susceptibility endpoints inconclusive and indicates that testing should be repeated	Innovotech Inc. (2025)

2.10 Ethical and quality considerations

The study was carried out in vitro using well-established reference strains, and it didn't involve any human participants, identifiable data, or tissue samples. Before diving into the practical work, we secured approval from the university along with the necessary risk assessments; you can find the approval reference as [INSERT ETHICS/PROJECT APPROVAL REFERENCE AND DATE], and the letter is included in Appendix A. Throughout the process, we adhered to aseptic techniques, containment protocols, and institutional waste management procedures. We also included blank, sterility, growth, and biofilm growth controls at every stage, and you can find the performance of these controls detailed in Chapter 3, regardless of whether they met the expected criteria.

3. Results

3.1 Antibiotic susceptibility screening and MIC determination

The diameters of the zones varied significantly between the two organisms (see Table 7 and Figure 1). For P. aeruginosa, the largest zone measured was for meropenem at 23.7 ± 1.53 mm, followed by gentamicin at 19.0 ± 1.00 mm, colistin sulphate at 14.3 ± 0.58 mm, and tetracycline at 11.7 ± 2.89 mm. Notably, penicillin, amoxycillin, chloramphenicol, and sulfamethoxazole didn't produce any measurable zones. On the other hand, S. epidermidis showed its largest zones with gentamicin at 17.0 ± 1.41 mm, sulfamethoxazole at 16.0 ± 0.00 mm, and linezolid at 14.0 ± 2.83 mm, while vancomycin had one of the smallest zones at 7.5 ± 0.71 mm. Interestingly, erythromycin displayed the widest range in replicates of any pairing, measuring 12.0 ± 8.49 mm (with values ranging from 6 to 18 mm). Vancomycin was included in the analysis due to its well-known effectiveness against staphylococci, despite its smaller zone size.

Table 7. The diameters of the zones of inhibition from the disc-diffusion screening were measured after incubating at 30 °C for 24 hours. We had three wells for P. aeruginosa and two for S. epidermidis. It's important to note that no clinical susceptibility category has been assigned, as explained in Section 2.4.

Antibiotic	**PA mean ± SD (mm)**	**SE mean ± SD (mm)**	**PA range**	**SE range**
Meropenem	23.7 ± 1.53	Not tested	22–25	—
Gentamicin	19.0 ± 1.00	17.0 ± 1.41	18–20	16–18
Colistin sulphate	14.3 ± 0.58	5.0 ± 1.41	14–15	4–6
Tetracycline	11.7 ± 2.89	8.0 ± 1.41	10–15	7–9
Sulfamethoxazole	0	16.0 ± 0.00	0	16–16
Linezolid	Not tested	14.0 ± 2.83	—	12–16
Erythromycin	Not tested	12.0 ± 8.49	—	6–18
Penicillin	0	8.5 ± 0.71	0	8–9
Vancomycin	Not tested	7.5 ± 0.71	—	7–8
Amoxycillin	0	Not tested	0	—
Chloramphenicol	0	Not tested	0	—

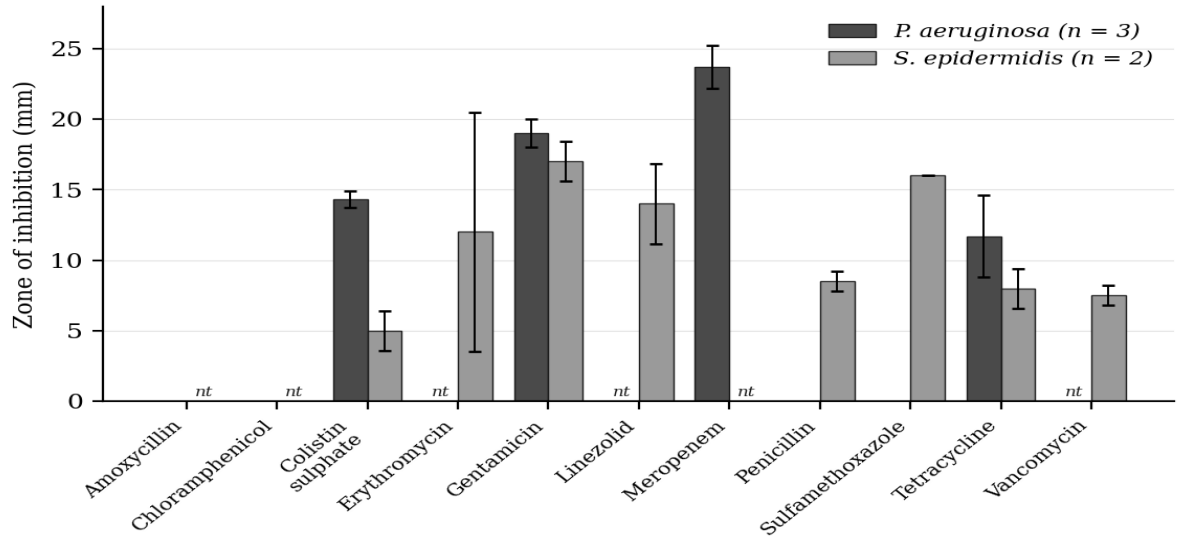


Figure 1. The diameters of the zones of inhibition for both organisms were measured across the antibiotic panel. The bars represent the mean values, with error bars indicating one standard deviation from the replicate measurements. Any agents that weren't tested against a specific organism are marked as "nt." If you see a zero-height bar, it means there was no measurable zone observed.

In Table 8, you'll find the MIC values. Both vancomycin and meropenem showed a value of 2 µg/mL, while the flavonoid MICs were significantly higher, ranging from one to three orders of magnitude. When it came to S. epidermidis, myricetin had the lowest MIC at 50 µg/mL, followed closely by curcumin at 64 µg/mL, then baicalein at 512 µg/mL, and finally orientin at 1024 µg/mL. For P. aeruginosa, curcumin and baicalein both recorded an MIC of 256 µg/mL. Unfortunately, neither orientin nor myricetin reached an endpoint within the tested range, so they were excluded from further studies on that particular organism.

Table 8. We determined the minimum inhibitory concentrations using broth microdilution with an MTT endpoint. If you see "Not reached," it means that we couldn't find any inhibitory concentration within the tested range.

Agent	***S. epidermidis* MIC (µg/mL)**	***P. aeruginosa* MIC (µg/mL)**
Vancomycin	2	Not tested
Meropenem	Not tested	2
Myricetin	50	Not reached
Curcumin	64	256
Baicalein	512	256
Orientin	1024	Not reached

3.2 Checkerboard interactions

We looked at three different combinations (see Table 9). The pairing of vancomycin and curcumin against S. epidermidis showed an FICI of 1.00, with both MICs halving, which perfectly fits the additive criterion. On the other hand, vancomycin combined with orientin yielded an FICI of 1.03: here, the orientin MIC dropped dramatically by 32 times, from 1024 to 32 µg/mL, while the vancomycin MIC stayed the same, nudging the index just above the

additive threshold. As for meropenem and curcumin, the FICI was 2.00, with no changes in either MIC. None of the combinations reached the synergy level, and thankfully, none showed signs of antagonism. Unfortunately, we couldn't get a reliable endpoint for the meropenem and baicalein combination, so we decided not to pursue that one further.

Table 9. Here are the results for the checkerboard fractional inhibitory concentration index. The FICI is calculated by taking the sum of each agent's MIC when used together and dividing it by the MIC of each agent on its own. All MIC values are measured in μg/mL. For reference, the interpretive bands can be found in Table 6.

Organism	Combination	Antibiotic MIC alone / combined	Flavonoid MIC alone / combined	FICI	Interpretation
S. epidermidis	Vancomycin + curcumin	2 / 1	64 / 32	1.00	Additive
S. epidermidis	Vancomycin + orientin	2 / 2	1024 / 32	1.03	Indifferent
P. aeruginosa	Meropenem + curcumin	2 / 2	256 / 256	2.00	Indifferent

3.3 Baseline biofilm-forming capacity

The optical density cut-off was set at 0.062, which established a strong-producer threshold of 0.248 (see Table 10). Both organisms surpassed this threshold on average. S. epidermidis showed a steady biomass production with a mean of 0.300 ± 0.024 and a median of 0.301. In contrast, P. aeruginosa displayed more variability, with a mean of 0.262 ± 0.139 and a median of 0.199. Notably, one well recorded a reading of 0.532, while five others ranged between 0.175 and 0.287, indicating a departure from normality (Anderson–Darling $p = 0.013$). When comparing the two organisms, there was no significant difference found (Mann–Whitney $W = 49$, $p = 0.128$).

Table 10. We assessed baseline biofilm formation using crystal violet staining at 550 nm, comparing the results to the optical density cut-off (ODc = 0.062; with a strong-producer threshold set at 4 × ODc = 0.248). Individual readings from each well are provided, allowing for a direct inspection of the distribution.

Organism	Individual OD550 readings	Mean ± SD	Median	Class
S. epidermidis	0.281, 0.306, 0.267, 0.314, 0.334, 0.296	0.300 ± 0.024	0.301	Strong
P. aeruginosa	0.181, 0.211, 0.186, 0.532, 0.287, 0.175	0.262 ± 0.139	0.199	Strong

3.4 Single-flavonoid effects on biofilm biomass

The effects observed were specific to both the compounds used and the organisms tested, and in some instances, they went against what we initially expected (see Table 11). Positive percentage values indicate a decrease in biomass compared to the untreated control from the same plate, while negative values show an increase in biomass.

When it came to S. epidermidis, baicalein was the standout performer, showing a clear concentration-dependent inhibition. The biomass dropped to 0.168 ± 0.006 at 2X and 0.157 ± 0.023 at X, which translates to reductions of 44.5% and 47.9%, respectively. Both of these results were significantly different from the control (ANOVA $F(4,11) = 33.51$, $p < 0.001$; Dunnett $p < 0.001$ for each), while the X/2 and X/4 concentrations did not show significant differences ($p = 0.343$ and $p = 0.772$). Myricetin also made an impact, reducing biomass by 31.9% at 2X, although the overall model just missed being significant ($F(4,11) = 3.19$, $p = 0.057$), with the planned 2X comparison showing $p = 0.035$. On the other hand, orientin didn't have any significant effect at any concentration ($F(4,11) = 1.27$, $p = 0.339$).

Curcumin's effect on S. epidermidis turned out to be quite surprising. At concentration X, the biomass measured 0.334 ± 0.003, compared to a control value of 0.302, marking a notable increase of 51.5%. This was the only comparison that showed a significant difference from the control (Welch $F(4,4.42) = 22.76$, $p = 0.003$; Games–Howell $p = 0.005$). Interestingly, all four concentrations of curcumin resulted in negative percentage values.

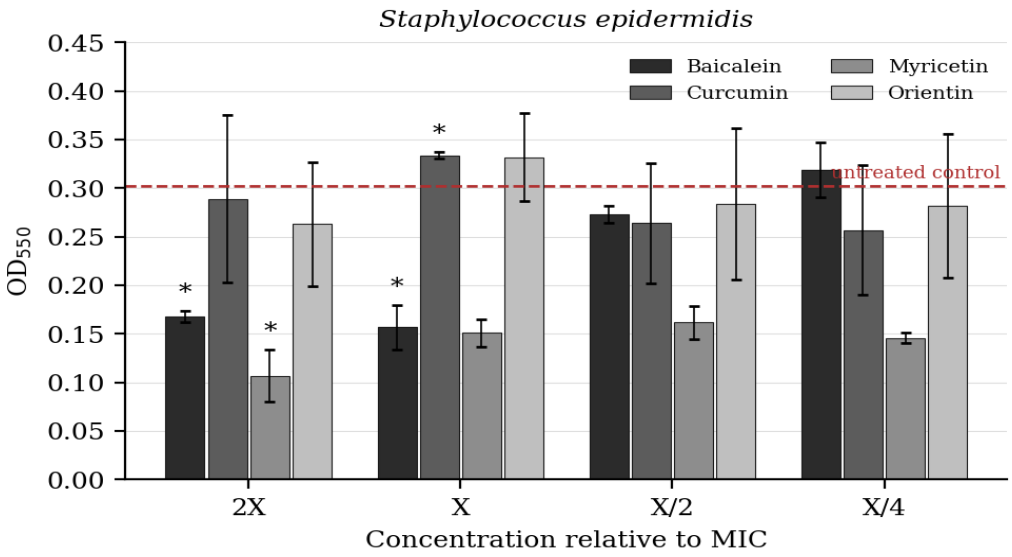


Figure 2. The biofilm biomass of S. epidermidis was assessed after treatment with various flavonoids at different concentrations, using OD550 as the measurement. The bars represent the average values along with one standard deviation from triplicate wells. The dashed line indicates the mean of the untreated control, which is 0.302. Asterisks highlight the comparisons that show a significant difference from the control group.

When it came to P. aeruginosa, the results took a different turn. Curcumin was effective in reducing biomass across all concentrations tested, showing a decrease of 59.4% at 2X and reaching up to 73.2% at X/2 (Welch $F(4,4.38) = 8.38$, $p = 0.026$). The concentrations of 2X, X/2, and X/4 all showed significant differences from the control (Games–Howell $p = 0.046$, 0.018, and 0.016 respectively), while X did not show a significant difference ($p = 0.178$). On

the other hand, Baicalein didn't have a notable impact (Kruskal–Wallis H(4) = 5.03, p = 0.284), and the variability at X/2 was greater than the mean itself (0.386 ± 0.427).

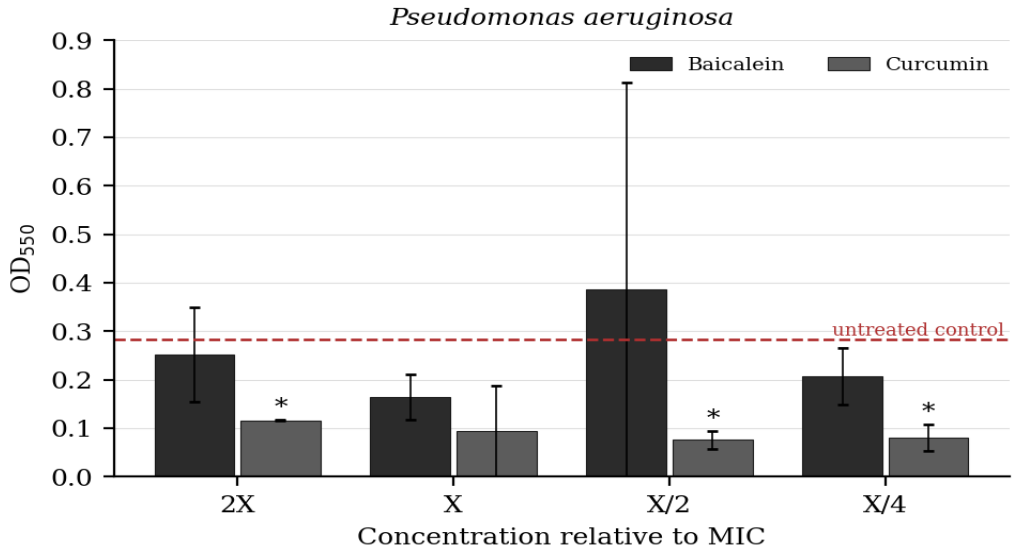


Figure 3. We measured the biofilm biomass of P. aeruginosa after treating it with curcumin and baicalein, using OD550 as our metric. The bars represent the average along with one standard deviation from three wells. The dashed line indicates the mean of the untreated control. It's worth noting that orientin and myricetin weren't tested on this organism, as we didn't reach a minimum inhibitory concentration (MIC) for them.

Table 11. We're looking at how single-flavonoids impact biofilm biomass across different concentrations. The percentage change is measured against the untreated control from the same plate—positive values indicate inhibition, while negative values show that the biomass has surpassed the control. You'll find the statistical details provided for each compound.

Organism	Flavonoid	Conc.	OD550 mean ± SD	Change (%)	Statistical outcome
S. epidermidis	Baicalein	2X	0.168 ± 0.006	+44.5	ANOVA F(4,11) = 33.51, p < 0.001. Dunnett: 2X p < 0.001, X p < 0.001, X/2 p = 0.343, X/4 p = 0.772
		X	0.157 ± 0.023	+47.9	
		X/2	0.273 ± 0.009	+9.8	
		X/4	0.319 ± 0.028	−5.5	
	Curcumin	2X	0.289 ± 0.086	−30.9	Welch F(4,4.42) = 22.76, p = 0.003. Games–Howell: X vs control p = 0.005; others not significant
		X	0.334 ± 0.003	−51.5	
		X/2	0.264 ± 0.062	−19.4	
		X/4	0.257 ± 0.067	−16.3	
	Myricetin	2X	0.107 ± 0.027	+31.9	ANOVA F(4,11) = 3.19, p = 0.057. Dunnett: 2X p = 0.035; others not significant

Organism	Flavonoid	Conc.	OD550 mean ± SD	Change (%)	Statistical outcome
		X	0.151 ± 0.014	+4.3	
		X/2	0.162 ± 0.017	−2.5	
		X/4	0.146 ± 0.005	+7.6	
	Orientin	2X	0.263 ± 0.064	−26.3	ANOVA F(4,11) = 1.27, p = 0.339. No comparison significant
		X	0.332 ± 0.045	−59.2	
		X/2	0.284 ± 0.078	−36.4	
		X/4	0.282 ± 0.074	−35.3	
P. aeruginosa	Baicalein	2X	0.252 ± 0.098	+11.4	Kruskal–Wallis H(4) = 5.03, p = 0.284
		X	0.164 ± 0.047	+42.5	
		X/2	0.386 ± 0.427	−35.7	
		X/4	0.207 ± 0.058	+27.4	
	Curcumin	2X	0.116 ± 0.001	+59.4	Welch F(4,4.38) = 8.38, p = 0.026. Games–Howell: 2X p = 0.046, X p = 0.178, X/2 p = 0.018, X/4 p = 0.016
		X	0.094 ± 0.093	+66.9	
		X/2	0.076 ± 0.018	+73.2	
		X/4	0.081 ± 0.027	+71.6	

3.5 Established-biofilm challenge: growth and metabolic activity

Both combinations showed a concentration-dependent reduction in OD650 (see Table 12 and Figure 4). For the vancomycin–curcumin combination, the values dropped from 0.372 at the lowest concentration to 0.112 at the highest, compared to a growth control reading of 0.364 (Welch $F(8,7.19) = 215.54$, $p < 0.001$). On the other hand, the vancomycin–orientin combination decreased from 0.545 to 0.123, but the overall test didn't reach significance (Kruskal–Wallis $H(8) = 11.84$, $p = 0.159$). This lack of significance can be attributed to notable variability among replicates at intermediate concentrations, where one series consistently showed higher readings than the other two.

At the peak concentration, the optical density for both combinations was nearly identical to that of the sterility control: 0.112 compared to 0.116 for curcumin, and 0.123 versus 0.125 for orientin. This indicates that the readings had hit the assay floor, rather than showing a slight

but detectable level of growth.

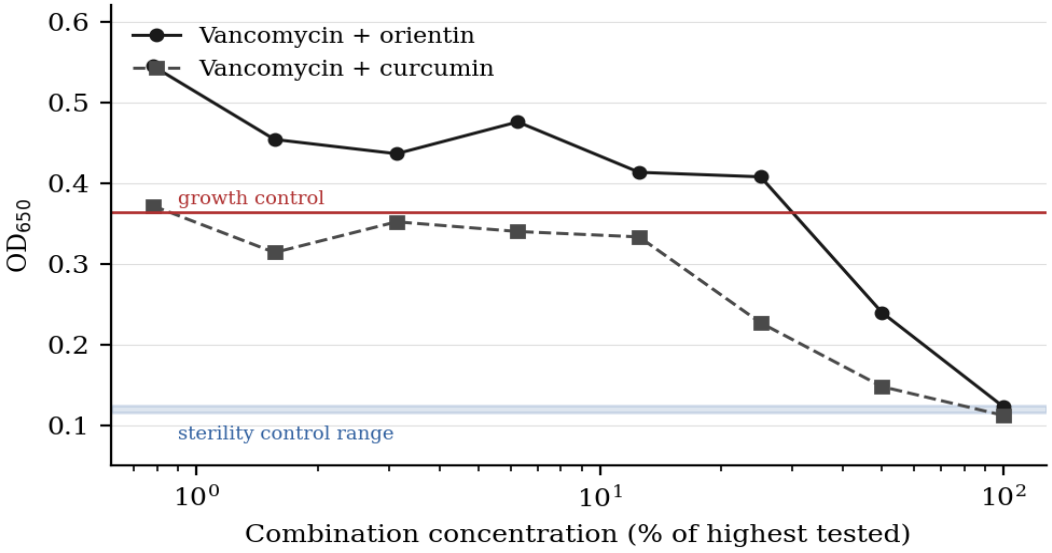


Figure 4. The optical density at 650 nm for the challenge-plate wells was measured across the combination concentration series and plotted on a logarithmic scale. You'll notice a solid horizontal line indicating the growth control, while the shaded area represents the range of the two sterility controls, which were tested without any inoculum.

MTT provided a qualitative endpoint (see Table 12). The only combination that stayed yellow was the highest concentration of vancomycin–orientin, which was 2048 µg/mL orientin paired with 4 µg/mL vancomycin. This suggests there was no noticeable reduction of the tetrazolium salt. In contrast, the vancomycin–curcumin well showed a slight blue hue, indicating some residual metabolic activity. Meanwhile, all the other wells turned purple.

3.6 Retained biofilm biomass after combination challenge

When we looked at the crystal violet staining of the challenge plate at 570 nm, we noticed that the reductions were mainly seen in the top two concentrations (check out Table 12 and Figure 5). The combination of vancomycin and curcumin led to a significant drop in retained biomass, with reductions of 43.8% and 45.5% at the 100% and 50% concentrations, respectively. These were the only concentrations that stood out compared to the control (ANOVA $F(8,18) = 8.23$, $p < 0.001$; Dunnett $p = 0.015$ and $p = 0.011$). On the other hand, vancomycin combined with orientin showed a biomass reduction of 33.8% and 34.4% at the same concentrations (Kruskal–Wallis $H(8) = 18.12$, $p = 0.020$). Below the 50% mark, the percentage reductions were minimal and often negative, dipping as low as −28.8%.

The sterility controls limit how we can interpret these values. In the wells that had the combinations but no intentional inoculum, we saw significant dye retention, with readings of 0.293 for vancomycin–curcumin and 0.383 for vancomycin–orientin. In comparison, the treated wells showed readings of 0.285 and 0.336 at the highest concentration. This means

that the amount of crystal violet retained at the top concentration was actually no higher than in the wells that didn't have any inoculum at all.

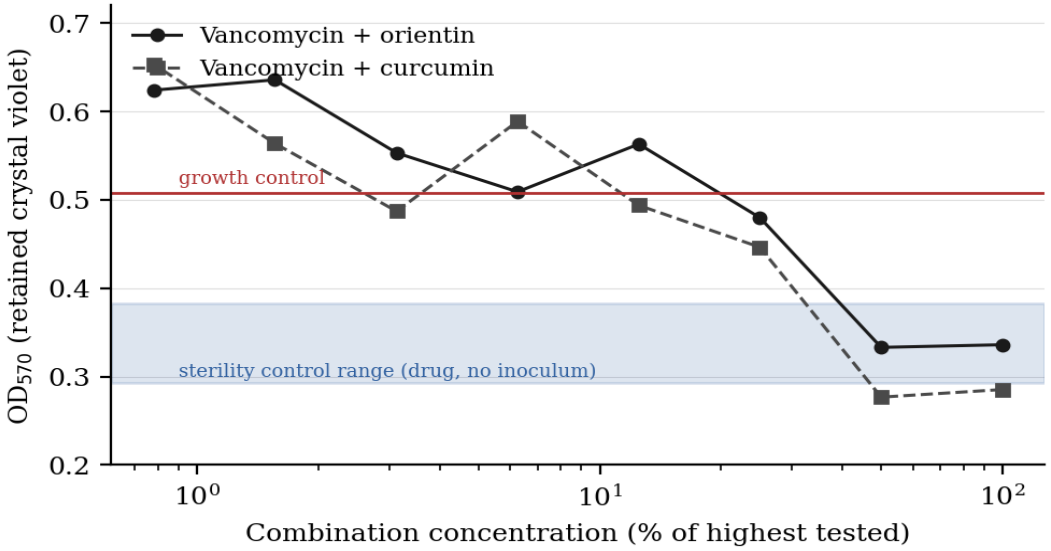


Figure 5. We measured the retained crystal violet at 570 nm across a series of different concentrations. The solid line represents the average growth control, while the shaded area indicates the range of the two sterility controls, which had the drug but no inoculum. The overlap we observed between the treated wells at the highest concentrations and the sterility control band is why we advise caution in Section 4.6.

Table 12. Here are the results for the established-biofilm challenge with both combinations. V+O refers to vancomycin paired with orientin, while V+C stands for vancomycin combined with curcumin. The percentage values in parentheses indicate the reductions in retained biomass compared to the growth control; if you see negative values, that means the biomass was actually higher than the control.

Conc.	OD650 V+O	OD650 V+C	OD570 V+O	OD570 V+C	MTT result
100%	0.123	0.112	0.336 (+33.8%)	0.285 (+43.8%)	V+O yellow, no growth; V+C slight blue
50%	0.240	0.148	0.333 (+34.4%)	0.277 (+45.5%)	Both purple
25%	0.408	0.227	0.480 (+5.5%)	0.446 (+12.0%)	Both purple
12.5%	0.413	0.333	0.563 (−11.0%)	0.494 (+2.7%)	Both purple
6.25%	0.476	0.340	0.509 (−0.3%)	0.589 (−16.0%)	Both purple
3.13%	0.436	0.352	0.553 (−9.0%)	0.487 (+4.0%)	Both purple
1.56%	0.454	0.314	0.636 (−25.3%)	0.564 (−11.1%)	Both purple
0.78%	0.545	0.372	0.624 (−23.0%)	0.653 (−28.8%)	Both purple
Growth control	0.364	0.364	0.507	0.507	Purple
Sterility control	0.125	0.116	0.383	0.293	Not applicable

3.7 Viable recovery, log10 reduction and control performance

The spot-titre enumeration showed almost no dynamic range, as illustrated in Table 13 and Figure 6. The challenge-plate counts ranged from 2.30 to 2.56 log10 CFU per peg for all concentrations of both combinations, compared to a growth control of 2.40, resulting in reductions between −0.16 and 0.10. For the recovery-plate counts, 25 out of 27 treated wells had a count of 1.322, while the growth control was at 1.419, leading to a maximum reduction of 0.10 log10. Importantly, there were no significant differences in the challenge datasets across the groups (H(8) = 11.55, p = 0.172; H(8) = 7.53, p = 0.481).

The recovery datasets showed some nominally significant tie-adjusted results: H(8) = 25.92, p = 0.001 for the vancomycin–orientin combination, and H(8) = 15.57, p = 0.049 for vancomycin–curcumin. However, this is largely due to the almost complete lack of variation; when we look at the unadjusted results, we see p = 0.462 and p = 0.521 for the same comparisons. It's important to note that these figures are included for thoroughness, not as proof of any treatment effect.

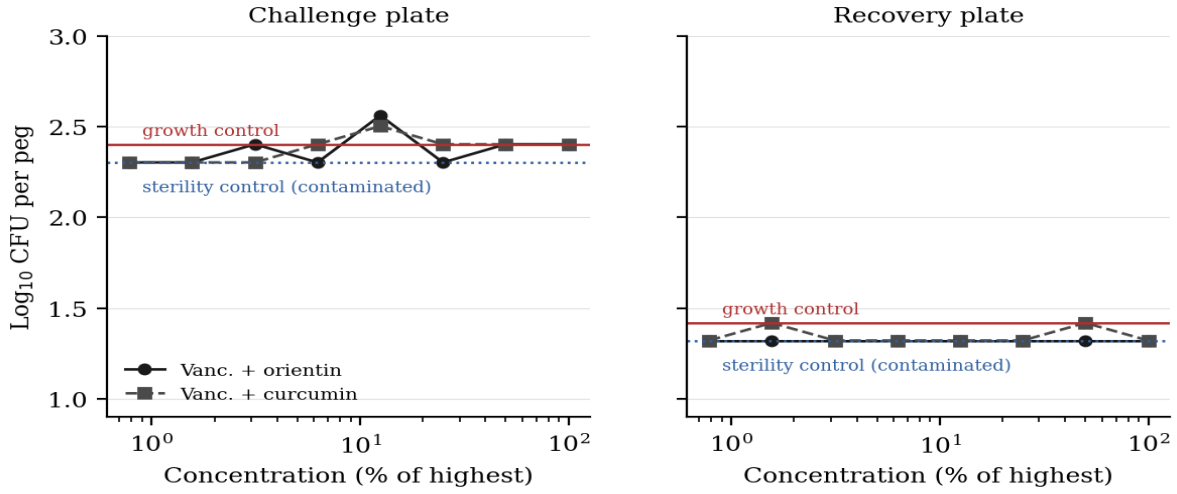


Figure 6. The recovery datasets showed some nominally significant tie-adjusted results: H(8) = 25.92, p = 0.001 for the vancomycin–orientin combination, and H(8) = 15.57, p = 0.049 for vancomycin–curcumin. However, this is largely due to the almost complete lack of variation; when we look at the unadjusted results, we see p = 0.462 and p = 0.521 for the same comparisons. It's important to note that these figures are included for thoroughness, not as proof of any treatment effect.

Table 13. Spot-titre viable counts and log10 reductions relative to the growth control. Reductions are shown for the vancomycin–orientin challenge series and for both recovery series; the vancomycin–curcumin challenge series ranged from 2.303 to 2.503 with reductions between −0.10 and 0.10.

Conc.	Challenge log10 V+O	Reduction	Recovery log10 V+O	Recovery log10 V+C	Reduction
100%	2.403	0.00	1.322	1.322	0.10
50%	2.403	0.00	1.322	1.419	0.00
25%	2.303	0.10	1.322	1.322	0.10

Conc.	Challenge log10 V+O	Reduction	Recovery log10 V+O	Recovery log10 V+C	Reduction
12.5%	2.562	−0.16	1.322	1.322	0.10
6.25%	2.303	0.10	1.322	1.322	0.10
3.13%	2.403	0.00	1.322	1.322	0.10
1.56%	2.303	0.10	1.322	1.419	0.00
0.78%	2.303	0.10	1.322	1.322	0.10
Growth control	2.403	—	1.419	1.419	—

The performance control details are laid out in Table 14, and it turns out we didn't meet one of the criteria. The sterility control wells, which had the highest concentration of the combination without any inoculum, showed colonies on spotting from both plates. According to the criterion in Table 6, any growth in these wells makes the susceptibility endpoints unclear. Additionally, the growth control counts were quite low, averaging just 1.33 colonies per spot, which means that the treated wells, growth controls, and sterility controls all fell within the same tight range. Unfortunately, none of the concentrations managed to completely eliminate growth after recovery, so we couldn't determine a minimum biofilm eradication concentration.

Table 14. Performance of assay controls against their prespecified criteria. Two entries are reported as not met, and their consequences for interpretation are examined in Sections 4.6 and 4.7.

Control	Purpose	Criterion	Observed
Blank (biofilm assay)	Establish background and ODc	Low, consistent optical density	Met; ODc = 0.062
Growth control (GC)	Confirm organism viability	Growth present	Met, but only 1.33 colonies per spot
Biofilm growth control (BGC)	Confirm biofilm formed on pegs	OD650 above 0.1	Met; mean OD650 = 0.366
Sterility control (SC)	Confirm absence of contamination	Wells must remain clear	NOT MET; colonies recovered from both plates
Overall assay validity	Permit assignment of MBEC	All controls must pass	NOT MET; endpoints inconclusive

4. Discussion

4.1 Principal findings

This study looked into the effects of four flavonoids on two organisms known for forming biofilms, using a sequential design that ended with a challenge involving established biofilms. Four key findings emerged from the research. First, the activity of the flavonoids varied significantly depending on the organism, and interestingly, for curcumin, the effects were actually opposite between the two species. None of the three combinations tested achieved the synergy threshold during the planktonic checkerboard tests. However, both combinations that moved on to the biofilm challenge showed concentration-dependent reductions in optical density and retained biomass at their highest concentrations. Importantly, though, these reductions didn’t lead to any decrease in recoverable viable cells, and the controls in the assay didn’t meet the necessary criteria for us to interpret the eradication endpoint at all.

The real value of this study isn't so much about finding an effective adjuvant, which it didn't manage to do, but rather in showing that three commonly used endpoints in antibiofilm research—optical density, crystal violet biomass, and viable recovery—actually produced different results when tested on the same biofilms under the same conditions.

4.2 Biofilm phenotype and the limits of threshold classification

Both organisms met the criteria for strong biofilm production, which confirms that the models were suitable for the questions we were exploring. However, the classification for P. aeruginosa is a bit less reliable than the summary statistic might imply. While its mean of 0.262 did surpass the 0.248 threshold, the median of 0.199 fell short. This mean was largely influenced by a single reading of 0.532, while five other readings ranged between 0.175 and 0.287. The deviation from normality (Anderson–Darling $p = 0.013$) supports this observation. So, the classification really hinges on which measure of central tendency you choose to use, putting us in a tricky spot with a threshold-based approach.

mmThis isn't just a trivial statistic. When it comes to pseudomonal biofilm formation in vitro, factors like the composition of the medium, shear forces, and surface characteristics play a crucial role, leading to noticeable differences between wells on the same plate. What this means in practice is that we should be careful when comparing the two organisms: the lack of a significant difference in baseline biomass ($p = 0.128$) could be just as much about the variability in the pseudomonal readings as it is about any real similarity.

4.3 Organism-dependent single-agent effects

The results from the single-agent tests were specific to both the compounds and the organisms involved, making them not easily applicable to other situations. Baicalein stood out as the only compound that showed a clear concentration-dependent inhibition against S. epidermidis, managing to reduce biomass by 44 to 48% at or above its minimum inhibitory concentration (MIC). This aligns with the antibiofilm properties previously noted for this flavone. However, the commonly cited mechanism—interference with quorum sensing in P. aeruginosa (Luo et al., 2016; Parra Rodríguez et al., 2026)—doesn't really explain its effects on a Gram-positive organism, which has a very different signaling system. Interestingly, baicalein didn't show any significant impact on P. aeruginosa in this study ($p = 0.284$), even though that's where its anti-quorum-sensing activity has been most thoroughly documented.

Orientin didn't show any significant effect against S. epidermidis at any concentration, which is the most straightforward test of a specific mechanism that's been published. Wang et al. (2021) pinpointed how orientin inhibits sortase A in S. aureus by identifying individual binding residues and showed that this leads to reduced biofilm formation. However, our current findings suggest that this effect didn't carry over to S. epidermidis in the conditions we tested. There are a few possible reasons for this: the two species have different surface protein profiles and matrix compositions; orientin's minimum inhibitory concentration (MIC) here was the highest of all the compounds we tested, sitting at 1024 µg/mL; and while sortase inhibition can lower adhesion, it doesn't necessarily mean there's a decrease in total stainable biomass, which is what crystal violet measures. So, this finding doesn't contradict Wang et al.; instead, it clarifies the limits of their results.

It's worth comparing the absolute MIC values we've obtained with what's already out there in the literature. For instance, Matilla-Cuenca et al. (2020) found that a Bap-expressing S. aureus strain had MICs of 375 µg/mL for curcumin and 1000 µg/mL for both myricetin and baicalein. In our current study with S. epidermidis, we observed significantly lower values for curcumin (64 µg/mL) and myricetin (50 µg/mL), while the value for baicalein (512 µg/mL) was quite similar. Some of this variation can be attributed to the different species and strains involved, as well as the use of the MTT endpoint in our study instead of visual turbidity. However, the trend remains consistent across the board: the MICs for flavonoids are generally one to three orders of magnitude higher than those of traditional antibiotics. This difference highlights why

we're looking at these compounds more as potential adjuvants rather than standalone treatments.

Myricetin led to a 31.9% reduction in biomass when used at double its minimum inhibitory concentration (MIC), with the overall model showing a p-value of 0.057. This aligns with previous findings that myricetin inhibits Bap-dependent staphylococcal biofilm formation by blocking amyloid assembly (Matilla-Cuenca et al., 2020). While the results are in line with what we expected, the lack of statistical significance might be more about the structure of the replicates discussed in Section 4.7 rather than indicating that there's no effect at all. There's one screening result that really stands out, especially when it comes to how we picked the comparator antibiotics. Vancomycin showed one of the smallest zones in the S. epidermidis panel, measuring just 7.5 ± 0.71 mm, but it also had a minimum inhibitory concentration (MIC) of 2 µg/mL, which is among the lowest recorded in this study. These two findings actually complement each other rather than contradict. You see, vancomycin is a large glycopeptide that doesn't diffuse well through agar, meaning its zone diameter tends to underestimate its effectiveness compared to smaller molecules. This issue is even more pronounced given the lower incubation temperature we used in this experiment. This discrepancy highlights why we treated the screen as more descriptive and why we chose vancomycin based on its pharmacological properties rather than just its zone size.

4.4 The curcumin reversal

Curcumin really stood out in this study. It boosted the biofilm biomass of S. epidermidis at its minimum inhibitory concentration (MIC) to 0.334, compared to a control of 0.302. At the same time, it slashed the biomass of P. aeruginosa by an impressive 59 to 73% across all concentrations tested. Such a significant shift in species like this definitely deserves some attention and explanation, rather than being brushed aside.

There are three interpretations that could fit here, and they don't necessarily rule each other out. The first one is about methodology. Curcumin has a vibrant color and doesn't dissolve well in water, while crystal violet measures the material that's stuck around rather than just the living cells. If the compound settles in the well or clings to the surface, it would get washed and stained along with the biofilm, which could lead to an inflated reading. The later challenge assay gives us solid proof that this isn't just a theory: the sterility control wells that had the drug but no inoculum still showed a crystal violet reading of 0.293, which is pretty much the same as the treated wells at the highest concentration.The second interpretation is biological.

The biofilm matrix acts like a sponge for hydrophobic substances (Flemming et al., 2023). This means that a hydrophobic compound might get trapped in the extracellular polymeric substances instead of making its way to the cells. Interestingly, in some organisms, sub-inhibitory stress can actually encourage the production of the matrix instead of hindering it. Either way, this would lead to an increase in measured biomass without suggesting that the antibacterial activity has failed.

The third point to consider is that the effect is both authentic and specific to different species. This is due to the unique envelope structures found in Gram-positive and Gram-negative organisms, which leads to varying levels of access to intracellular targets. To truly differentiate between these possibilities, we need to rely on viability-based measurements, something that crystal violet simply can't offer. What we can say for sure is that we shouldn't automatically assume curcumin is beneficial for all organisms, and using it in optical assays comes with a specific risk of producing artefacts, which has now been clearly demonstrated.

4.5 Planktonic interaction did not predict biofilm behaviour

None of the three combinations hit the synergy threshold. The vancomycin–curcumin pairing was additive at exactly 1.00, while vancomycin–orientin and meropenem–curcumin showed indifference at 1.03 and 2.00, respectively. Here are two key takeaways.

First, the result for vancomycin–orientin is more intriguing than its classification suggests. The MIC for orientin dropped dramatically—by 32 times—when combined, going from 1024 to 32 µg/mL, while the MIC for vancomycin stayed the same. Since the FICI is a sum, this asymmetry gets condensed into a single value that just barely crosses the additive threshold. A 32-fold decrease in the amount of natural product needed is exactly the kind of outcome an adjuvant strategy aims for, yet the index fails to recognize it as an interaction because the advantage went to the "wrong" partner. This is a known quirk of the metric, not a flaw in the data itself.

Second, and perhaps more importantly, the checkerboard results didn’t accurately predict biofilm behavior. The vancomycin–curcumin combination, which was the only additive pairing, showed the strongest optical suppression in the challenge assay. In contrast, vancomycin–orientin, labeled as indifferent, achieved complete MTT clearance. However, neither was able to eradicate the biofilm. This disconnect was anticipated in Section 1.10 and aligns with the recognized limitation of using a planktonic growth-inhibition endpoint for a

sessile population, where factors like diffusion, metabolic diversity, and persister subpopulations come into play (Ciofu et al., 2022).

4.6 Optical suppression was not eradication

One of the most significant findings is the disagreement between the endpoints in the established-biofilm challenge. When looking at the optical density data on its own, it seems promising: both combinations reduced OD650 in a concentration-dependent manner, with the highest concentrations showing values that were nearly identical to those of sterile wells. The crystal violet data backs this up, showing reductions of 34 to 46% at the top two concentrations. However, when it comes to viable recovery, the results were quite disappointing, with log10 reductions never surpassing 0.10, which falls short of the three-log reduction typically seen as a sign of effective antibiofilm activity.

We need to break down two different explanations for the optical results. First, turbidity at 650 nm decreases when cells are killed. However, it also drops when cells are simply stopped from growing out from the peg. Interestingly, the readings hit the lowest point, matching the sterility control value in both scenarios. This alignment with a no-cell control, rather than just a low positive value, suggests that we're seeing a lack of planktonic growth instead of the complete eradication of the biofilm on the peg surface.

The results from the crystal violet assay show similar limitations. At the highest concentration, the amount of retained dye in the treated wells, which measured 0.285 and 0.336, was not significantly different from the sterility control wells that contained the drug but no inoculum, which had readings of 0.293 and 0.383. When comparing to an untreated growth control, any reduction observed can't be confidently linked to a loss of biofilm biomass, especially since a well with just the drug produced comparable results.

There's another internal inconsistency that really highlights this point. The biofilm growth control showed a mean OD650 of 0.366, which is well above the 0.1 threshold that suggests biofilm formation on the pegs. However, this same control only produced 1.33 colonies per spot when we counted them. If we had a well-established biofilm at this dilution, we would expect to see a solid growth. So, the optical and viable measurements from the control don't match up, suggesting that the differences we're seeing aren't just limited to the treated wells but are actually a characteristic of how the assay was conducted.

The only exception here is the MTT result at the highest concentration of vancomycin–orientin, which is the sole well that didn't show any tetrazolium reduction. It's important to note that metabolic quiescence doesn't equate to death; the recovery well still produced colonies. So, it's more accurate to interpret this as a suppression of metabolic activity rather than complete eradication.

When you look at the three endpoints together, it becomes clear that a less thorough design might have mistakenly reported them as aligned, when in fact they systematically disagree, leaning towards an overestimation of efficacy. If the study had only measured OD650 and crystal violet, it would have led to the conclusion that both combinations significantly diminished established biofilms. However, the recovery data indicates that this conclusion lacks solid support.

4.7 Limitations and confidence in the data

There are a few limitations that affect what can be claimed, and the biggest issue revolves around the eradication endpoint itself. The sterility control wells showed growth on both the challenge and recovery plates. According to the protocol, this means that the susceptibility endpoints are inconclusive, and further testing is necessary (Innovotech Inc., 2025). So, the right takeaway here isn't that eradication didn't work, but rather that it couldn't be measured: with contaminants showing up at about one colony per spot and treated wells showing the same, it's impossible to tell surviving biofilm cells apart from the contamination.

There's another issue that complicates things. The growth control pegs only produced 1.33 colonies per spot, while we would expect a fully developed biofilm at this dilution to show dense growth. It's unclear if this is due to sparse colonization of the pegs or if the recovery process during sonication was ineffective. Either way, the assay had almost no dynamic range, meaning we wouldn't have been able to detect a three-log reduction even if it had happened. We need to tackle both of these problems before we can revisit the question of eradication.

The way the replicate structure is set up really limits inference on a broader scale. The triplicates were all wells from the same plate instead of being independent experiments, which means the p-values we see only reflect variation within that plate and can't be applied more generally. This leads to two important consequences. First, results that don't quite hit significance, like myricetin at $p = 0.057$, might be more about this structure than a true lack of effect. Second, just because some results do reach significance doesn't mean they're guaranteed to be reproducible. The recovery-plate tie-adjusted values highlight this issue from

the other side, showing nominal significance from data where 25 out of 27 observations were the same.

We ran susceptibility screening at 30 °C instead of the standard 35 ± 2 °C, which means we can't directly compare the zone diameters to clinical breakpoints, and that's why we didn't assign a susceptibility category. For the baseline biofilm, we measured it at 550 nm, while the challenge-plate biomass was measured at 570 nm, so we can't compare the absolute optical densities between these two assays, although comparisons within each assay remain valid. Lastly, it's important to note that the organisms we used were laboratory reference strains, and the wound isolates behave differently in terms of biofilm formation and resistance factors (Ghasemian et al., 2023; Dinić et al., 2024), so we can't really apply these findings to a clinical context based on this evidence.

4.8 Implications and future work

The results don’t back up the hypothesis as it was presented. There wasn’t any combination that showed synergy, and none were able to eliminate an established biofilm in a way that could be clearly understood.

They back three specific recommendations. First, the eradication assay needs to be repeated using strict aseptic techniques and verified sterility controls. It’s important to confirm peg colonization before any challenges and to select dilutions that ensure control counts are in a countable range rather than teetering on the edge of zero. Second, endpoints should be reported in parallel instead of selectively, as the differences noted here would go unnoticed in a study that only measures biomass. Lastly, curcumin should be reevaluated using a viability-based method. This is crucial not only because its species reversal is fascinating from a mechanistic standpoint, but also because the risk of artifacts in optical assays has now been shown to be a real concern, not just a theory.

There are two leads that definitely deserve some extra attention. First off, the 32-fold drop in the orientin MIC when paired with vancomycin is intriguing. Even though it shows indifference according to the FICI, this kind of asymmetric potentiation is exactly what an adjuvant program would want to explore further. Ideally, we’d want to use a method that breaks down each agent’s contribution individually. On another note, the consistent suppression of P. aeruginosa biofilm biomass by curcumin, ranging from 59 to 73% across all four concentrations, really needs to be confirmed against a viability endpoint specifically for that organism. This is particularly important because it stands out as the only single-agent

effect in this dataset that was both significant and consistent throughout the entire concentration range.

5. Conclusions

We looked at four different flavonoids to see how they fared against Staphylococcus epidermidis and Pseudomonas aeruginosa, using a step-by-step approach that started with susceptibility screening and ended with a biofilm challenge. Both bacteria showed a strong ability to form biofilms. The effectiveness of the flavonoids varied depending on the organism: baicalein managed to cut down staphylococcal biofilm biomass by 44 to 48% at or above its minimum inhibitory concentration (MIC), while curcumin was able to reduce pseudomonal biomass by 59 to 73% across all tested concentrations. However, curcumin had the opposite effect on staphylococci, actually increasing their biomass. Interestingly, orientin didn’t show any measurable impact on S. epidermidis, even though it has been effective against S. aureus in other studies.

None of the antibiotic-flavonoid combinations showed any signs of synergy. The combination of vancomycin and curcumin produced an additive effect, while the other pairs didn’t really interact at all. Interestingly, the minimum inhibitory concentration (MIC) of orientin dropped by 32 times when paired with vancomycin, which is a unique effect that the fractional inhibitory concentration index doesn’t quite reflect.

When tested against established biofilms, both combinations were effective in reducing optical density and maintaining biomass at the highest concentrations. However, neither achieved a viable log10 reduction greater than 0.10. Since the sterility controls showed growth, the eradication endpoint failed to meet the validity criteria outlined in the protocol, meaning we couldn't assign a minimum biofilm eradication concentration. The clear takeaway here is that we couldn't demonstrate eradication, and based on this dataset, it simply wasn't detectable.

The main takeaway here is about the methods used. When we looked at optical density, crystal violet biomass, and viable recovery in the same biofilms, we noticed they all showed a consistent trend of exaggerating effectiveness. So, if you come across studies claiming antibiofilm activity based on just one optical measurement, it's wise to take those findings with a grain of caution.

References

Each entry listed below has been carefully checked against the publisher or PubMed record, ensuring accuracy in details like authors, year, title, journal, volume, issue, pagination, and DOI. The references follow Harvard style as outlined in Cite Them Right. This list pertains solely to Chapter 1 and will expand as we finalize the other chapters.

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Appendices

Appendix A. Ethics and project approval

Please include the Middlesex University ethics/project approval letter here. The module handbook specifies that this approval letter should be part of the appendices.

Appendix B. Raw antibiotic susceptibility data

Here are the individual zone diameters measured in millimeters for various antibiotics against P. aeruginosa and S. epidermidis. For P. aeruginosa, the results are as follows: penicillin showed no effect (0, 0, 0); gentamicin had diameters of 20, 18, and 19; amoxycillin also showed no effect (0, 0, 0); tetracycline had 15, 10, and 10; chloramphenicol was ineffective (0, 0, 0); sulfamethoxazole showed no effect (0, 0, 0); meropenem had diameters of 25, 22, and 24; and colistin sulphate measured 15, 14, and 14. For S. epidermidis, the results were: penicillin with diameters of 9 and 8; gentamicin at 16 and 18; tetracycline at 7 and 9; colistin sulphate at 4 and 6; vancomycin at 7 and 8; sulfamethoxazole at 16 and 16; erythromycin at 6 and 18; and linezolid at 12 and 16.

Appendix C. Raw baseline biofilm data

Here are the optical density readings at 550 nm for the individual wells: For S. epidermidis, the values are 0.281, 0.306, 0.267, 0.314, 0.334, and 0.296. Meanwhile, for P. aeruginosa, the readings are 0.181, 0.211, 0.186, 0.532, 0.287, and 0.175. The optical density cut-off (ODc) is set at 0.062, and the threshold for strong producers is 0.248.

Appendix D. Raw challenge and recovery plate data

Challenge-plate OD650 and OD570 readings, and spot-titre colony counts for both combinations, are reproduced in Tables 12 and 13 of Chapter 3. Individual replicate readings underlying those means are held in the accompanying analysis workbook.

Appendix E. Statistical decision record

For each dataset, residual normality was assessed by the Anderson–Darling test and variance homogeneity by Levene's test, and the test applied was determined by the outcome. One-way ANOVA with Dunnett comparisons was used where both assumptions held: baicalein and myricetin against S. epidermidis, orientin against S. epidermidis, and OD570 for vancomycin–

curcumin. Welch's ANOVA with Games–Howell comparisons was used where variances were unequal but residuals normal: curcumin against both organisms and OD650 for vancomycin–curcumin. Kruskal–Wallis was used where residuals departed from normality: baicalein against P. aeruginosa, OD650 and OD570 for vancomycin–orientin, and all four spot-titre datasets. Mann–Whitney was used for the two-organism baseline comparison. This record is provided so that test selection can be audited independently of the results it produced.